Agentic AI in mathematical research

A brief disclaimer

I am not an expert in this field. I have been exploring it out of curiosity and to stay up to date.

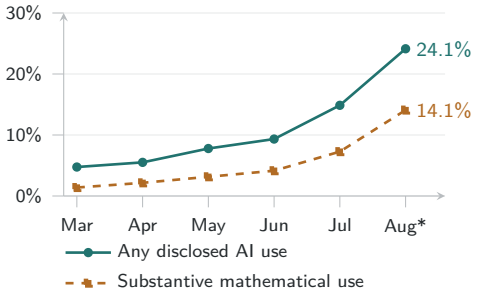
This talk neither argues for nor argues against using AI to write papers.

- A brief history and overview of mathematical research with LLMs, focusing on agentic approaches.
- What the literature tells us about successful approaches and their limitations.
- A case study where I build an AI research team to write a paper.

AI use and mathematical output

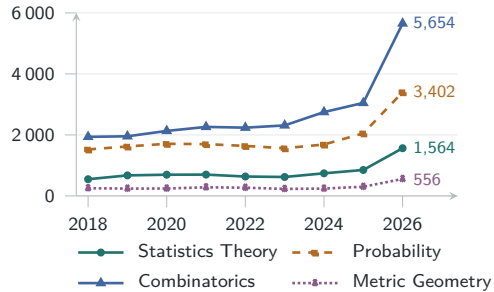
AI disclosures on arXiv

Share of math-related submissions, 2026



Submissions by field

New preprints, January–August each year

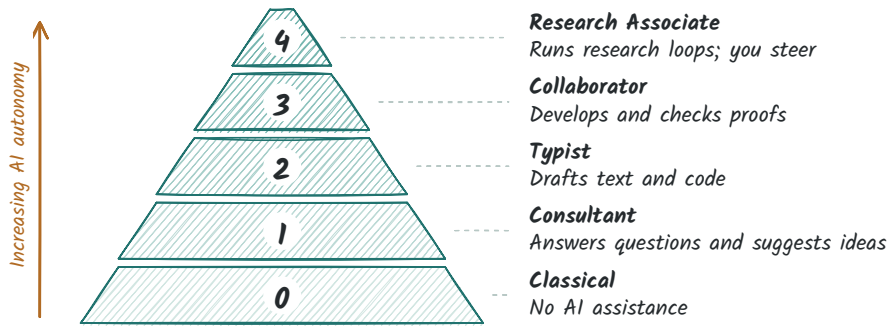


Left: 3,575 of 32,944 preprints disclosed AI use. *August covers 1–20.
Right: category counts for published arXiv papers.

Jin, Ke & Sui (2026), Table 1 Explore arXiv Math Pulse GitHub source.

(How) have you used AI in your research?

Which is the highest level you have used?



Adapted from Zimmer et al. (2026), *The Agentic Researcher*, Section 2.

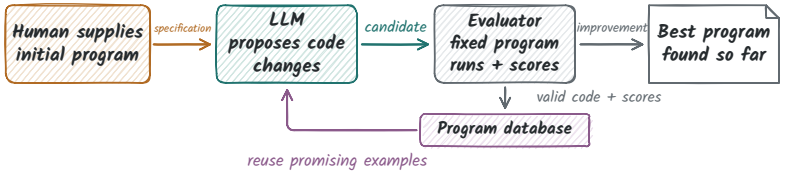
A short history of math research with LLMs

Agentic research systems over time

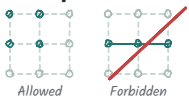
A brief history, with a focus on mathematics.

FunSearch Romera-Paredes et al., *Nature* (2024)

An evolutionary approach with fixed evaluation criteria implemented in computer programs.



Larger cap sets

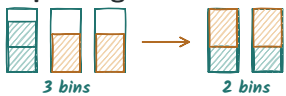


496 → 512

points in 8 dimensions

No three distinct points on a line in $\mathbb{F}_3^n$.
A new construction, not a proof of maximality.

Online bin-packing heuristics



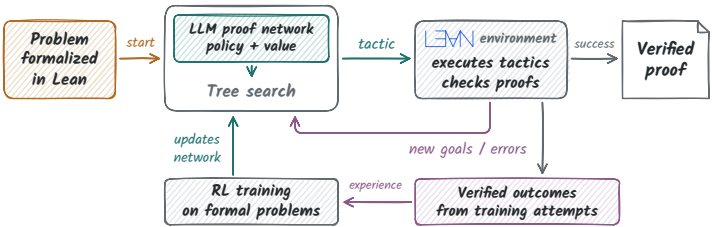
Use as few fixed-capacity bins as possible; place each item on arrival.

AlphaEvolve (2025) later extended this approach: 4×4 complex matrix multiplication with 48 scalar multiplications, versus Strassen's 49.

Workflow: Fig. 1; results: Fig. 4 and Table 1. Romera-Paredes et al. (2024).

AlphaProof Hubert et al., *Nature* (2026)

Another way to verify and evaluate mathematics is through formal verification in Lean.



Model: a 3-billion-parameter encoder–decoder transformer, pretrained on code and mathematics, then fine-tuned on human Lean proofs from Mathlib.

Search: the model reads Lean goals and hypotheses, proposes Lean tactics, and Lean returns new proof states.

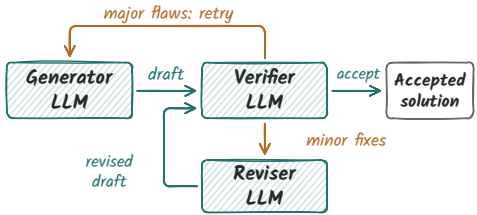
Learning: the model learns from proof attempts through reinforcement learning.

IMO 2024: 28/42 points, silver-medal level (AlphaProof: 3 problems; AlphaGeometry 2: 1).
Figs. 1–2; AlphaProof and Training sections. *Nature* 651, 607–613 (2026).

Aletheia Feng et al., arXiv (2026)

We can also use an independent LLM for verification. This does not guarantee correctness.

Gemini Deep Think agents



LLM judgment; no formal guarantee (Fig. 1).

Reported mathematical results

- Erdős: 2 full + 2 partial new solutions from 700 problems, assessed by experts.

And other mathematical achievements.

Other LLM research systems

The AI Scientist *Nature* (2026)
Ran ML experiments and wrote a paper that passed a workshop's first review round.

Google Co-Scientist *Nature* (2026)
Agents proposed and refined leukaemia drug combinations, validated in laboratory experiments.

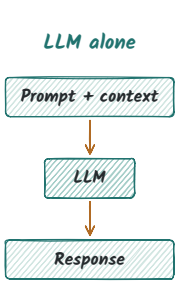
Virtual Lab *Nature* (2025)
LLM agents designed SARS-CoV-2 nanobodies with binding confirmed in laboratory tests.

OpenAI: Navier–Stokes report (2026)
Different approaches explored with 10,000 agents and human steering.

Agentic systems

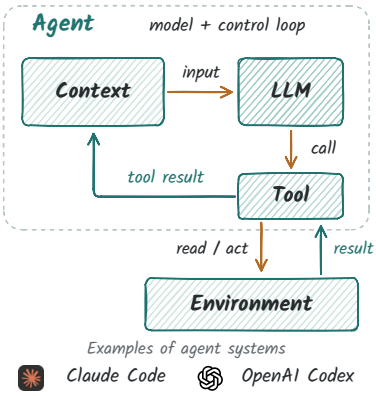
From LLMs to research teams

Environment, context, tools and agents



$$Y = f_\theta(p, c; \xi)$$

p: prompt; c: context
θ: fixed model
ξ: randomness



Environment
The workspace an agent can access: files, data, code and external services.

Context
The information given to the LLM. This includesinstructions, messages and tool results.

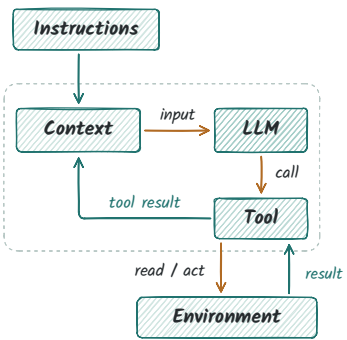
Tool
An interface to observe or change the environment, e.g. PDF-reader or Python ...

Agent
An LLM-based system that uses tools and chooses its next step from results.

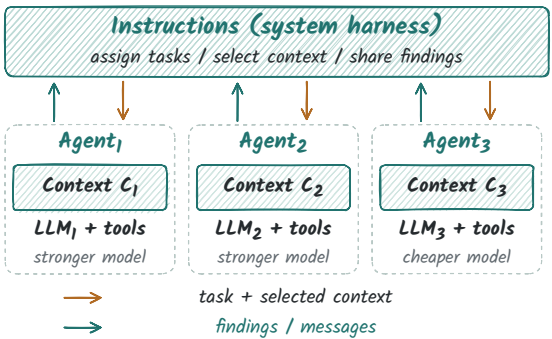
Based on OpenAI's Codex harness documentation and Claude Code's agentic loop.

Instructions and multi-agent systems

One agent



Several agents



Each agent has its own context. Instructions specify its task and how it should work.

Same or different models. Sources: ReAct (Yao et al., 2022); AutoGen (Wu et al., 2023).

Why manage context and communication?

An LLM produces text $Y = f_\theta(p, c; \xi)$ whose distribution depends strongly on the context.

1. Keep context focused Liu et al. (2024); Modarressi et al. (2025)
Give each agent the information it needs; long histories can obscure relevant details.
This is called: context distraction and context clash.

2. Give review a fresh context Sharma et al. (2023); Dhuliawala et al. (2023)
Review the draft and criteria without the writer's history; verify independently.
This is called: context poisoning and context confusion.

3. Match the model to the task Ong et al., *RouteLLM* (2024)
changing between stronger and cheaper models can improve the balance of cost and quality.

4. Share findings and avoid duplicate work Du et al. (2023); Wang et al. (2025)
Share useful results and redirect work that adds no new information.

5. Explore distinct approaches independently Zhu et al. (2026)
Especially important for mathematics since different starting points can lead to differnt proof strategies.

Effects depend on the task and models. Fresh-context review is a design inference from bias and verification studies.

Agentic research: a worked example

What do we need?

- An agent system: in my case, Codex.
- A precise mathematical problem statement.
- Relevant related work: definitions, known results and useful methods.
- `instruction.md`: the task, constraints and expected output.
- `skills.md`: an index of reusable methods and checks.
- `AGENTS.md`: standing project instructions for the agents.

Here, `instruction.md` and `skills.md` are our project documents.
Each actual Codex skill has its own `SKILL.md`.

Step 1: Defining the problem

A **simple point process** X is a random set of distinct points in $\mathbb{R}^d$, finite in bounded regions.

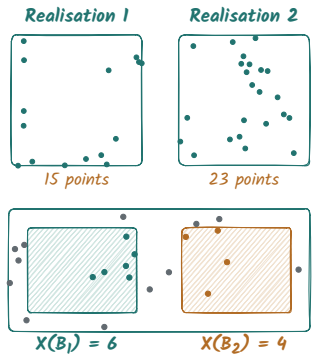
Its count in a bounded Borel region B is $X(B) = \#(X \cap B)$.

Stationarity means that shifting all points leaves the distribution unchanged:

$$X + t \overset{d}{=} X \quad \text{for every } t \in \mathbb{R}^d.$$

Homogeneous Poisson process, intensity $\lambda > 0$

- We demand that
$$X(B) \sim \text{Poisson}(\lambda|B|), \quad \mathbb{E}[X(B)] = \lambda|B|.$$
- Counts in disjoint regions are independent.
- Therefore, for $B_1 \cap B_2 = \emptyset$,
$$\mathbb{E}[X(B_1)X(B_2)] = \mathbb{E}[X(B_1)]\mathbb{E}[X(B_2)].$$



Baddeley, Sections 1.4–1.6. Pictures show simulated Poisson realisations in two dimensions.

Step 1: Defining the problem

1. **Count distinct tuples.** For $n \geq 1$, define the **factorial moment measure**
$$\alpha^{(n)}(A) := \mathbb{E}\Big[\sum_{(x_1,\dots,x_n)\in X^n_{\neq}} \mathbf{1}_A(x_1,\dots,x_n)\Big], \quad A \subseteq (\mathbb{R}^d)^n \text{ Borel.}$$
Here $X^n_{\neq}$ contains all ordered n-tuples of distinct points of X.

2. **Use disjoint regions.** For pairwise disjoint $B_1,\dots,B_n$,
$$\alpha^{(n)}(B_1 \times \cdots \times B_n) = \mathbb{E}[X(B_1)\cdots X(B_n)].$$
One point from each region gives distinct points. No independence is needed.

3. **Introduce a density.** When it exists, the **product density** $\rho^{(n)}$ satisfies
$$\alpha^{(n)}(A) = \int_A \rho^{(n)}(x_1,\dots,x_n)\,\mathrm{d}x_1\cdots\mathrm{d}x_n.$$

4. **Read it locally.** For small disjoint regions around x_1, x_2,
$$\mathbb{E}[X(B_1)X(B_2)] \;\approx\; \rho^{(2)}(x_1,x_2)\,|B_1|\,|B_2|,$$
provided $\rho^{(2)}$ varies little across $B_1 \times B_2$.

Assume bounded Borel windows and finite local moments.
Baddeley, Section 2.3; Last and Penrose, Section 4.4.

Step 1: Defining the Problem

For a homogeneous Poisson process, independence gives

$$\rho^{(2)}(x_1, x_2) = \lambda^2, \qquad \rho^{(n)}(x_1, \ldots, x_n) = \lambda^n.$$

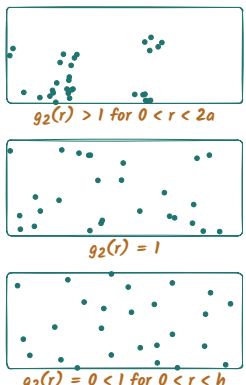
For a stationary process with intensity $\rho^{(1)} = \lambda > 0$, compare its product density with this reference:

$$g_n(x_1, \ldots, x_n) = \frac{\rho^{(n)}(x_1, \ldots, x_n)}{\lambda^n}.$$

This is the *n*-**point correlation function**.

Compared with Poisson at the same intensity:

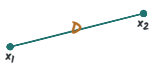
$g_n > 1$: more such tuples.
$g_n < 1$: fewer such tuples.
$g_n = 1$: the same local tuple density.



Simulated stationary processes, all with $\lambda = 20$; $r = \|x_1 - x_2\|$, $a = 0.10$, $h = 0.11$.
Comparisons are local; $g_2 = 1$ alone does not imply Poisson. Baddeley, Section 2.3; spatstat documentation.

Step 1: the precise problem

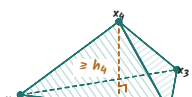
Fix $M, \lambda_0, h_3, h_4 > 0$.
Find one stationary, simple point process X on $\mathbb{R}^3$ with intensity λ_0, finite local moments through order four and product densities.



Distance D



Area A



Volume V

For some constants $C_2, C_3, C_4 > 0$, require simultaneously:

$$g_2(x_1, x_2) = \frac{C_2}{D}, \qquad g_3(x_1, x_2, x_3) = \frac{C_3}{A}, \qquad g_4(x_1, x_2, x_3, x_4) = \frac{C_4}{V}.$$

Almost everywhere on nondegenerate configurations of diameter $\leq M$, with every triangle altitude $\geq h_3$ and every tetrahedron altitude $\geq h_4$.

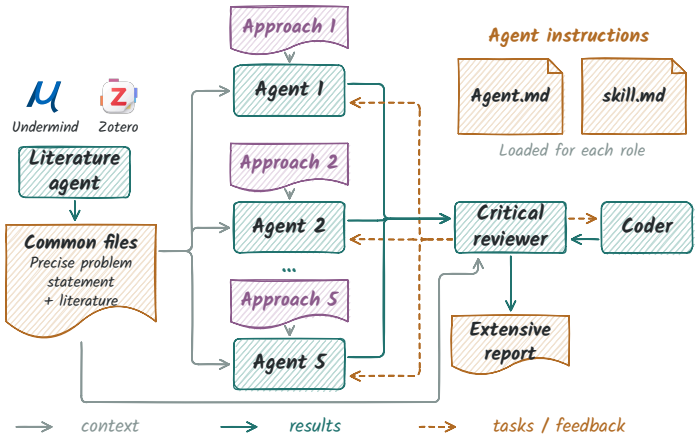
The conjecture is motivated by observed spatial correlations in earthquake catalogues.

Kagan and Knopoff (1980); Kagan (1981): three-point and four-point studies.

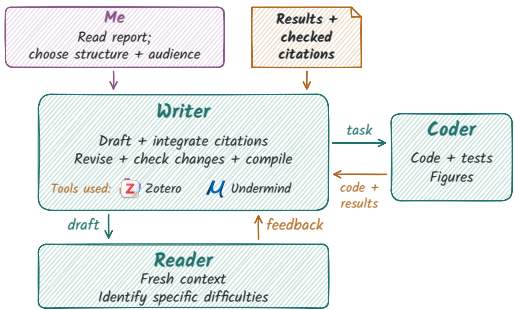
Step 2: configuration/agent files and execution

Research approaches

1. Poisson cluster process
2. Cox process
3. Hawkes process
4. An approach of the agent's choice
5. Prove non-existence



Step 3: Writing the paper



	Calculations	Writing
Tokens	132.5M	5.78M
Time	3 h 47 min	—
Price	$78.81	$9.44

M = million tokens, including cached input.
API-equivalent prices in USD.
Writing time unavailable in the saved summary.

Results and personal feedback

Poisson cluster model

Homogeneous Poisson parents Π each generate 0–3 children, with jointly chosen offsets C_z:

$$X = \bigcup_{z \in \Pi} \big(\{z\} \cup (z + C_z) \big).$$

C_z contains only child offsets. The parent is retained, giving 1–4 points per cluster.

How many points? The solution gives a probability distribution over cluster sizes $j = 1, \ldots, 4$.

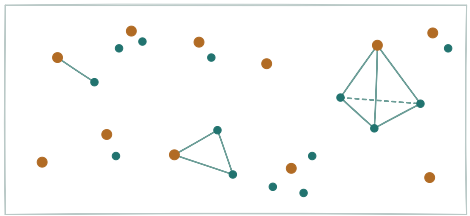
Where are they placed? For $j \geq 2$, the normalised density

$$\frac{F_j(z, x_1, \ldots, x_{j-1})}{Z_j}$$

describes how the $j - 1$ children are jointly distributed around the parent z.

Personal feedback

Aspect	Feedback
Mathematical correctness	Positive
Research and process documentation	Positive
Code writing	Positive
Text writing	Medium
Paper structuring	Negative
Mathematical explanation	Medium



Future of agentic research

What I expect to change

- Stronger models can increasingly plan, divide work into tasks, use tools and delegate to subagents.
- For a precise mathematical problem, one initial prompt may increasingly suffice. This can still mean substantial autonomous work, with no guarantee of success.
- I also expect explanations to become less sloppy and more useful for learning.

Where a custom harness still helps

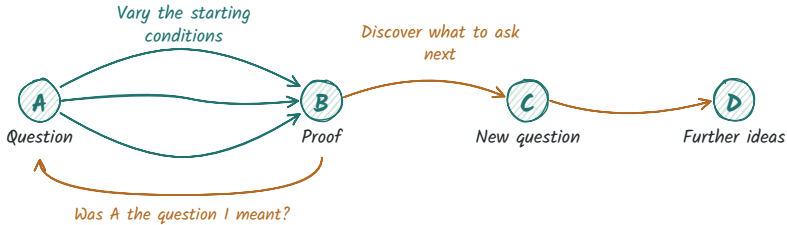
- Keep the exact target, hypotheses and criteria for success fixed.
- Require separate review and formal verification.
- Keep human checkpoints for direction and interpretation.
- Record results, failed approaches, sources and dependencies.
- Set compute budgets, stopping rules and model choice.
- When writing papers with AI, set your preferences for writing and illustration styles.

A reported example. For its October release, OpenAI says almost all results came from one prompt to one agent. Some may have needed retries; the full workflow has not been independently reproduced.

Scientific American, October 2026; OpenAI release details.

Personal take

Doing math this way can be still exciting.
Faster proofs let me explore more, different starting conditions let me explore different paths.
Still, something is missing when I do not discover the ideas myself.



Reaching B only helps me if I understand the mathematics.
Reaching B in different ways lets me think about where to go next i.e. C.

My concern as a PhD student: outsourcing too much may bypass the learning.
A PhD is still an education. Producing results and developing understanding can come apart.

Related: *A Severe Misalignment of AI in Mathematics* (joint declaration, 11 September 2026).

Thank you!

Questions and discussion

Slides and, later, the experiments:

github.com/MathisRost/agentic-ai-math

